Set up the Development Environment

Milestone 1 - Activity 1.3

1. Overview

Activity 1.3 prepares your computer so ComicCraft can be built and run. You install Python and pip, create an isolated virtual environment, install the required libraries, run the server for the first time and set up the project folder structure. It comes after choosing the AI models (Activity 1.1) and defining the architecture (Activity 1.2).

2. Steps at a glance

Step	Task	Why
1	Install Python and pip	Python runs the project; pip installs the libraries.
2	Create a virtual environment	Keeps project packages separate from system-wide Python.
3	Install required libraries	Gets FastAPI, AI libraries, PDF and image tools.
4	Run the server	Confirms the setup works.
5	Set up project structure	Organises code, templates and generated files.

3. Step 1 - Install Python and pip

Make sure **Python** is installed along with **pip**, which manages the project's dependencies. Check both in a terminal:

```
python --version
pip --version
```

If Python is missing, install it from python.org. Tick the option to add Python to PATH during installation on Windows.

4. Step 2 - Create a virtual environment

A virtual environment ensures that project-specific packages do not conflict with system-wide installations. Create and activate it:

```
python -m venv comiccraft-env

comiccraft-env\Scripts\activate      # Windows
source comiccraft-env/bin/activate     # macOS/Linux
```

Check: After activation, your terminal prompt shows `(comiccraft-env)` at the start. Always activate it before working on the project.

5. Step 3 - Install required libraries

```
pip install fastapi uvicorn jinja2 python-multipart google-generativeai \
    diffusers transformers fpdf Pillow accelerate
```

Library	Used for
fastapi	Web framework: routes, form and JSON handling.
uvicorn	ASGI server that runs the FastAPI app.
jinja2	HTML templates (index, comic preview, export success).
python-multipart	Lets FastAPI read HTML form data via Form(...).
google-generativeai	Calls Gemini Flash and Gemini Pro.
diffusers	Hugging Face library that runs Stable Diffusion.
transformers	Model components required by Diffusers.
fpdf	Builds the comic PDF.
Pillow	Image handling and saving.
accelerate	Helps load and run large models efficiently.

Diffusers needs **PyTorch** to run. It is normally installed along with these packages; if you get an error about torch, install it from pytorch.org.

6. Step 4 - Run the server

```
uvicorn app.main:app --reload
```

The `--reload` option restarts the server automatically whenever you change the code. Then open:

- `http://127.0.0.1:8000` - the ComicCraft app.
- `http://127.0.0.1:8000/docs` - FastAPI's interactive API documentation for testing endpoints.

7. Step 5 - Set up the project structure

Create the initial project directory structure for ComicCraft. The project document shows this as a screenshot; based on the files and folders named in the document, it looks like this **(example layout - yours may differ slightly)**:

```
ComicCraft/
|-- app/
|   |-- main.py             # FastAPI app entry point
|   |-- routes.py           # all routes
|   |-- gemini_flash.py     # generate_outline()
|   |-- gemini_pro.py       # generate_story()
|   |-- image_generator.py  # generate_image()
|   |-- layout_builder.py   # build_comic_layout()
|   `-- exporters.py        # save_pdf()
|-- templates/
|   |-- index.html
|   |-- comic_preview.html
|   `-- export_success.html
|-- static/
|   |-- panels/             # generated images
|   `-- exports/            # generated PDFs
|-- requirements.txt
`-- .env                    # API keys (do not share)
```

8. Verification checklist

- python and pip show version numbers.
- Virtual environment is created and activated.
- All libraries installed without errors (pip list shows them).
- uvicorn starts without errors and the homepage opens at 127.0.0.1:8000.
- /docs page opens.
- Project folders and files are created as planned.

9. Common problems (general tips)

Problem	Fix
'python' is not recognized	Reinstall Python with PATH option, or use py or python3 instead.
Cannot activate venv on Windows PowerShell	Use Command Prompt, or allow scripts for the current session in PowerShell.
Import error for a library	Check that the venv is activated, then run pip install again.
Uvicorn: Could not import module app.main	Run the command from the project root, where the app folder is.
Port 8000 already in use	Stop the other process or start with --port 8001.